# PRODUCT DESIGN & ENGINEERING **PORTFOLIO**

Up to date portfolio (Online)

Timothée MEYER

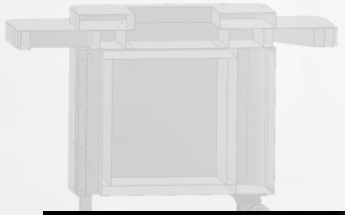
# ABOUT **ME**

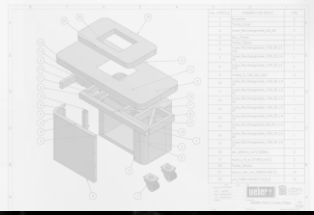
My name is **Timothée Meyer**, I am a French 5th year engineering student in **Design and Product Engineering** at UTBM. Passionate about *creating products* that serve a true purpose, *researching for innovation*, and working with partners *internationally*, I am currently looking for an internship that would allow me to flourish in at least one of these areas. The following projects illustrate the **last year and a half** of my engineering journey. Some were group projects within the framework of some of my courses, while others were purely self-initiative.

I consider myself as a person **driven by innovation and design**, with a great desire to launch my own products someday.




# GROUP PROJECTS





# WEBER® BOREAL

**[ Project Leader ]** : complete development of a party cooler fridge for the brand Weber® : benchmark, function, design, CAE and DFMA.

***{ 6 MONTHS }***







# MIRAGE 2000D 1:9

**[ 3D Parts & Paint Manager ]** : 1:9 scale hand modeled Dassault Mirage 2000D, delivered to the French DGA. Centered around perceived quality (surfaces, play, paint…)

***{ 6 MONTHS }***







DGA

DIRECTION GÉNÉRALE DE L'ARMEMENT



VANMOOF

# E-Bike SUV Frame

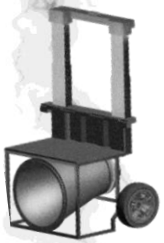
**[ Project Leader ]** : Merging of an SUV's design language and that of an E-Bike, with the manufacturing constraints of a bicycle, aided by an industry expert and rendered in a mix of hand-made sketches and computer renders.

***{ 2 WEEKS }***

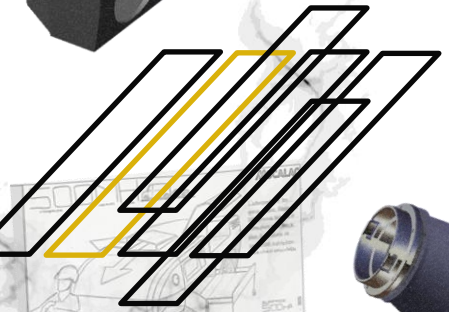


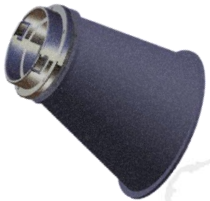


















**[ CAD Designer ]** : Development of a heating solution inspired by Mecalac for construction work application. This portable suitcase supplies warm air via a sheath from a central heater.

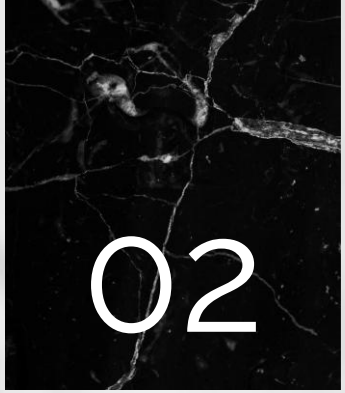
***{ 6 MONTHS }***

**MECALAC**

CR9

MECALAC





02

# DESIGN

CATIA

blender

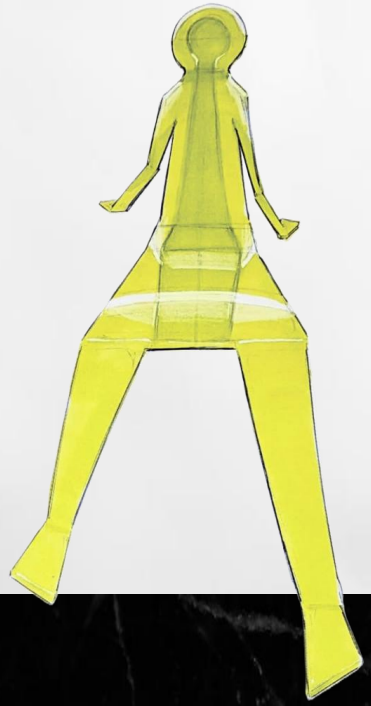
KRITA

# FIGU CHAIR

**[ Solo Product Designer ]** : Sketching, ideation, design and architecture choice for manufacturing and packaging, of a living room chair mimicking a sunbathing posture.

***{ 2 MONTHS }***







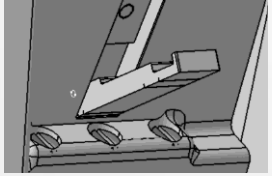


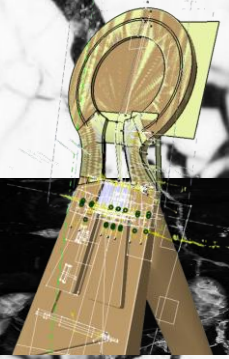
# INSPIRATION

Inspired by the relaxed pose of person sat at the beach sunbathing, with limbs serving as the chair's legs, and a crevasse running down the middle of the chair creating one continuous line from one leg to its symmetry...





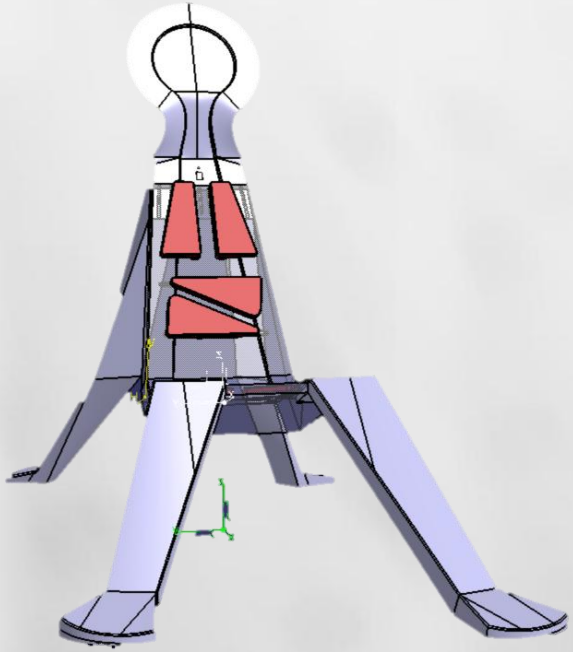


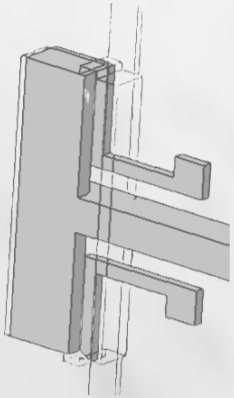


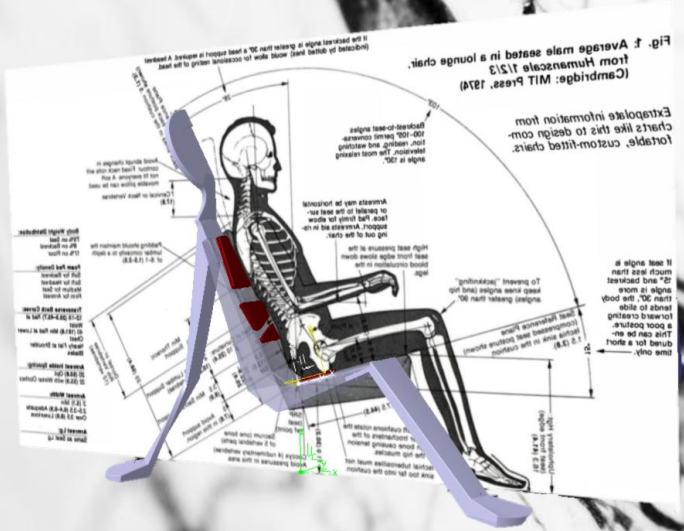
Fi
gu

# CAD DESIGN

Catia V5 GSD design and assembly architecture with screws to create an easily packageable and ergonomic chair, with removable cushion powered by elastic deformation to enable greater customization, and a rear hook to attach clothes and bags.









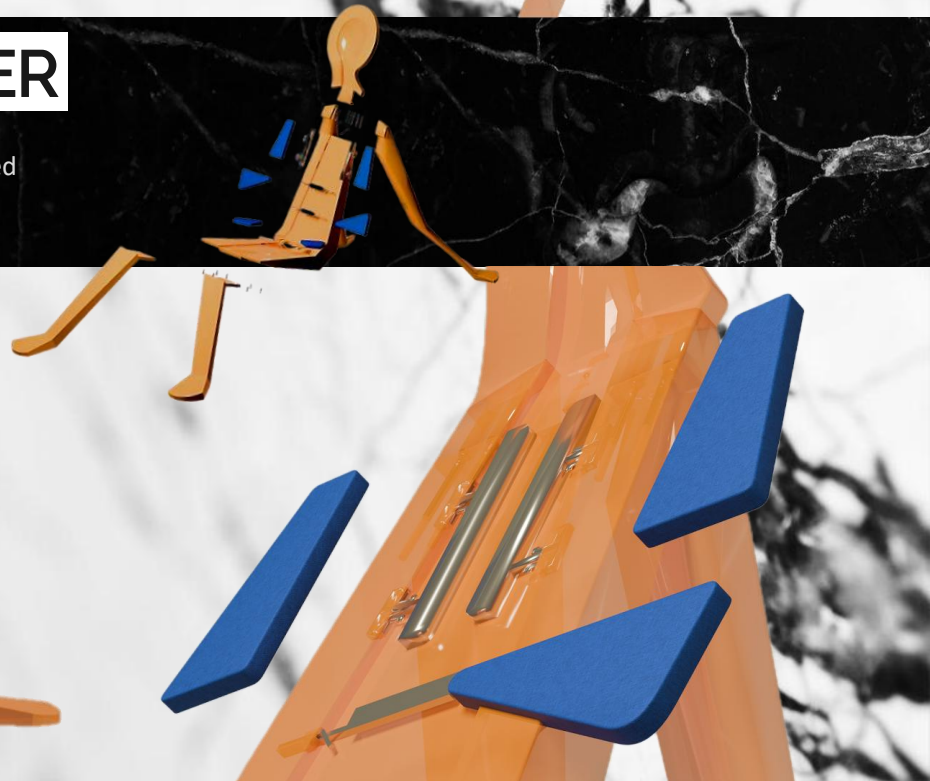




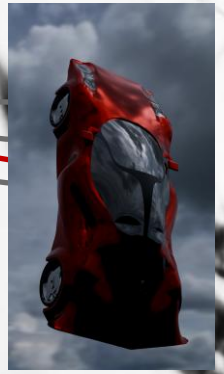
# RENDER

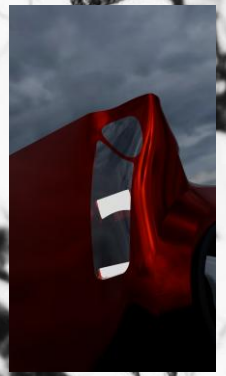
Materials, texture and lighting applied in Blender for the final render.

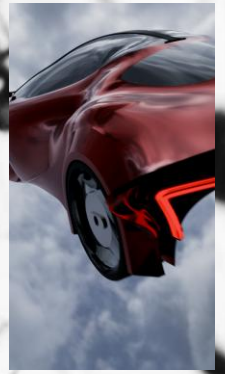












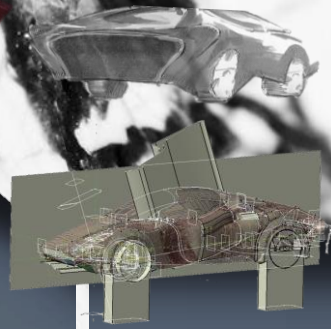




# NURBS Car surface

**[ Solo Designer ]** : Catia V5 GSD NURBS exercise to practice working with complex surfaces from sketches, with a sports car as a subject.
***{ 1 week }***

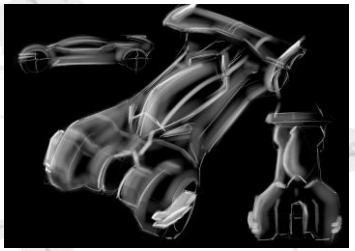


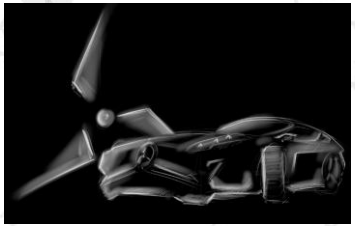


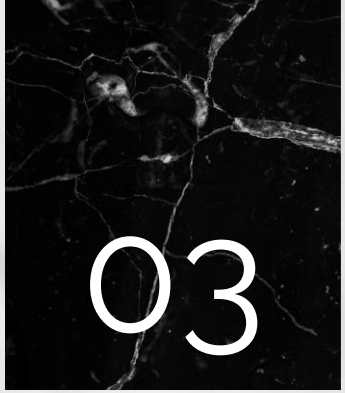


**[ Solo Sketcher ]** : Acquisition of product sketching skills for different applications : rapid strokes to detailed renders depending on the final use case.

# Sketching







# 03

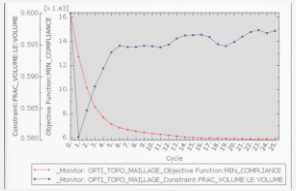
# CAE

CATIA

SIMULIA ABAQUS

TOSCA STRUCTURE

Microsoft Teams



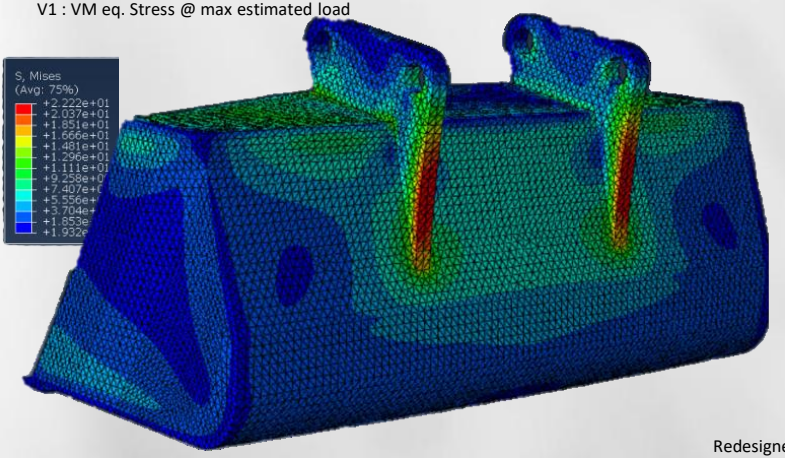

# Extraction Bucket Topological Optimization

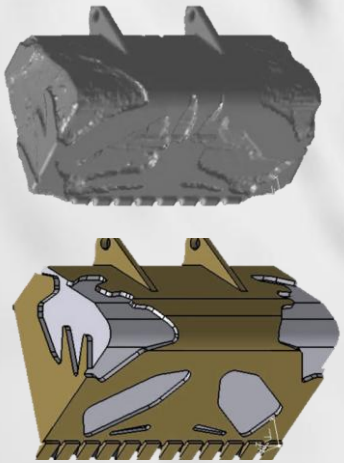
**[ CAE Modeler & Analyst ]** : Definition, calculations, modelling, simulation, optimization, redesign and final simulation of an extraction bucket, to reduce weight while maintaining structural integrity with a safety coefficient. Through a compliance minimising objective function, I redesigned a manufacturable bucket using the STL provided by the optimisation.

***{ 1 WEEK }***

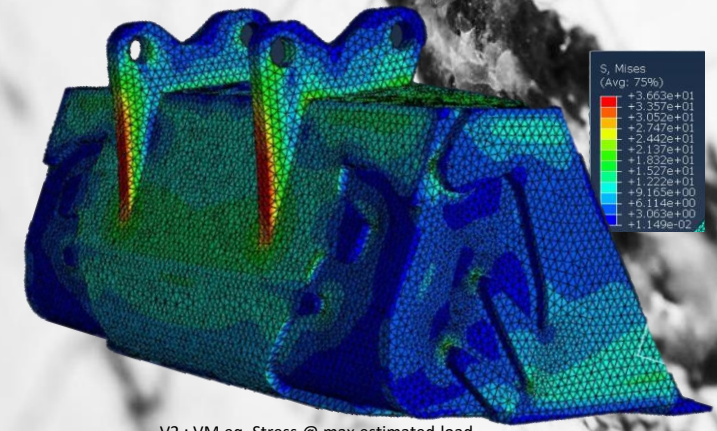


V1 : VM eq. Stress @ max estimated load

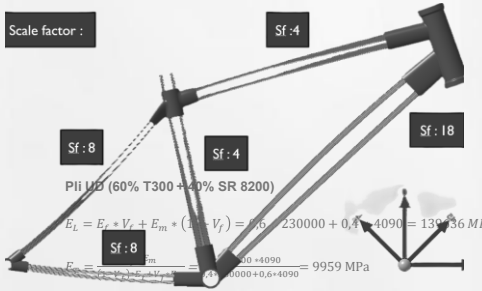


Redesigned manufacturable excavation bucket in Catia V5




V2 : VM eq. Stress @ max estimated load




$$E_L = E_f * V_f + E_m * (1 - V_f) = 0,6 * 230000 + 0,4 * 4090 = 139636\ MPa$$

$$E_T = \frac{E_f * E_m}{E_f * V_m + E_m * V_f} = \frac{230000 * 4090}{0,4 * 230000 + 0,6 * 4090} = 9959\ \text{MPa}$$

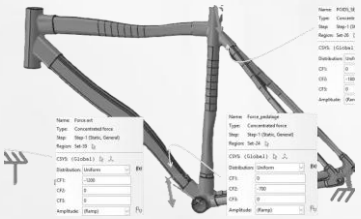
$$G_{LT} = \frac{G_f * G_m}{(1 - V_f) * G_f + V_f * G_m} = \frac{1460 * 88461}{0,4 * 88461 + 0,6 * 1460} = 3562\ \text{MPa}$$

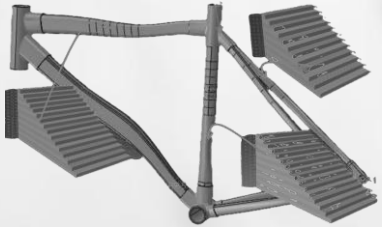
$$\vartheta_{LT} = \vartheta_f * V_f + \vartheta_m * (1 - V_f) = 0,6 * 0,3 + 0,4 * 0,4 = 0,34$$

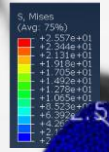
$$\vartheta_{TL} = \vartheta_{LT} * \frac{E_T}{E_L} = 0,34 * \frac{9959}{139636} = 0,024$$

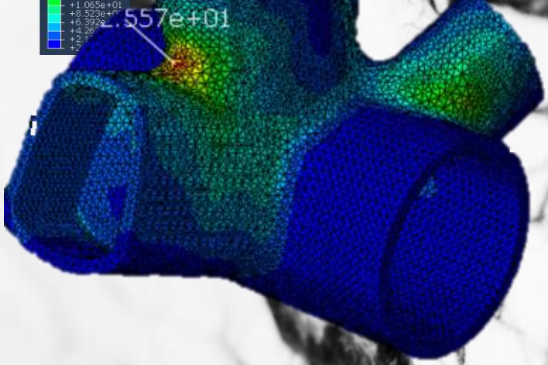
$$\rho_{composite} = \rho_f * V_f + \rho_m * (1 - V_f) = 0,6 * 1760 + 0,4 * 1175 = 1526\ \frac{kg}{m^3}$$

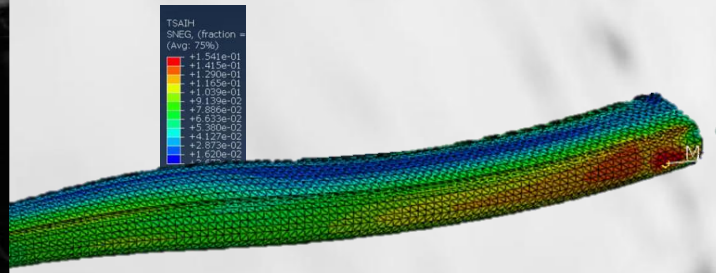
| Angle | Proportion |
|---|---|
| 0° | 50% |
| 45° | 12.5% |
| 90° | 25% |

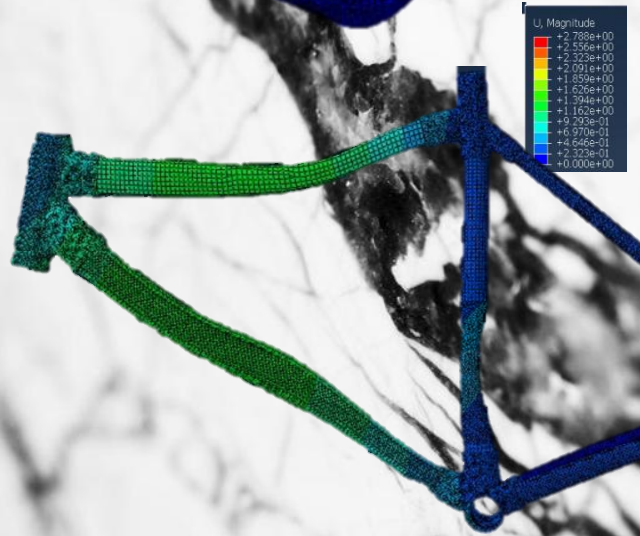







# Composite Bike Frame

**[CAE Modeler & Analyst ]** : Design of a carbon fibre bike frame in CATIA V5 and Abaqus, varying the number of plies around the saddle to make the ride smoother, and simulating a maximum load case on the tubes (VM eq. stress) and the tubes (TSAIH criterion)

***{ 3 WEEKS }***


04
ERGONOMICS

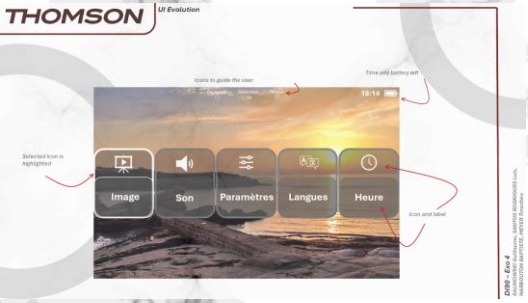
**[ UX/UI Designer ]** : Formal re-design of a Thomson DVD player based on Bastien & Scapin Heuristics, normalised user tests, as well as current design trends in Thomson products; to offer a product up to today's standards and expectations.

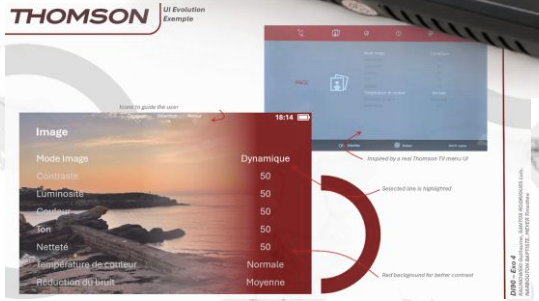
# THOMSON HMI Redesign

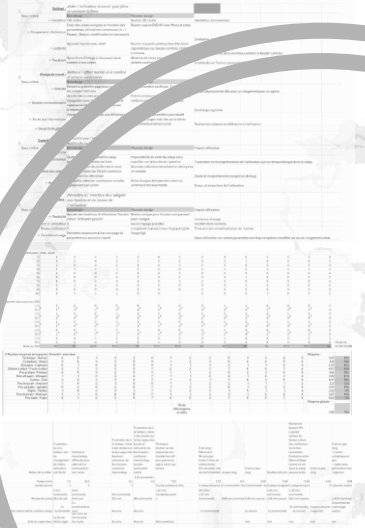








# CONTACT

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Up to date portfolio (Online)

Up to date CV (Online)